Data Management, Winter 2026/27 · Group project

From raw race results to a FAIR, quality-assured knowledge graph

Formula 1 World Championship results 1950–2025

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FAIR assessment and metadata design
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OVERVI
EW

Agenda & Roadmap

01

Executive Summary

Project objectives and core system goals.

02

Project and Data Architecture

Event-driven pipeline triggers and components.

03

FAIR Assessment

Comparative FAIR Assessment and decisions.

04

Data Quality

Validation framework and anomaly handling.

05

Monitoring & Ops

Latency tracking and error logging.

06

Conclusion

Key takeaways and future roadmap.

CORE GOALS

Project Objectives

01 **End-to-End Medallion Architecture**
Automated incremental pipeline from Bronze to Gold via PySpark and Parquet.

02 **Hybrid Vector and Graph RAG**
Combined vector embeddings with Neo4j property graphs for context retrieval.

03 **Parallel ML Engine Registry**
Automated training across 6 baselines including XGBoost, LightGBM, and GraphSAGE.

04 **Interactive Streamlit Analytics Suite**
High-performance local web app exposing data pipeline, models, and LLM assistant.

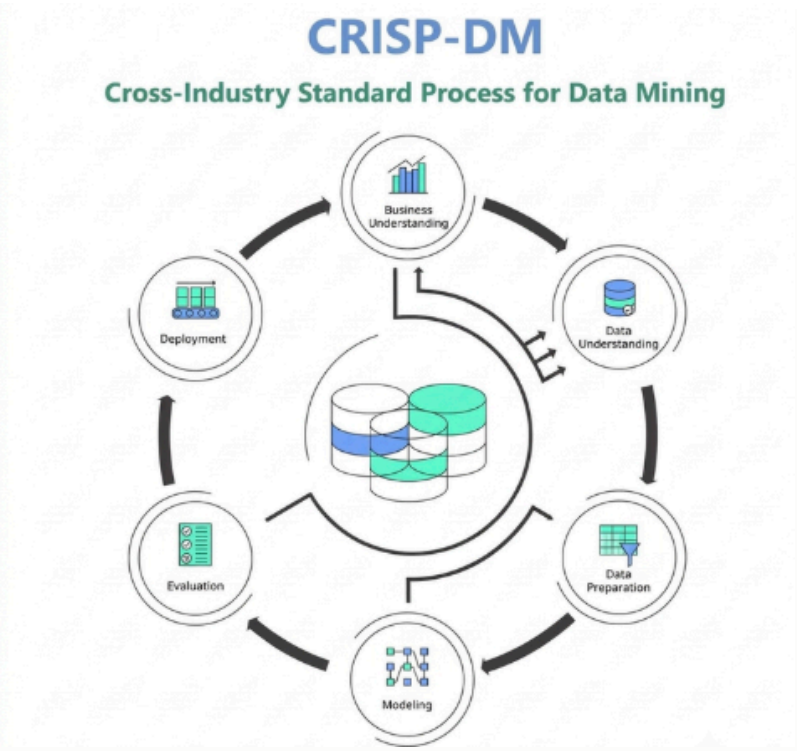
SECTION 01

Project and Data Architecture

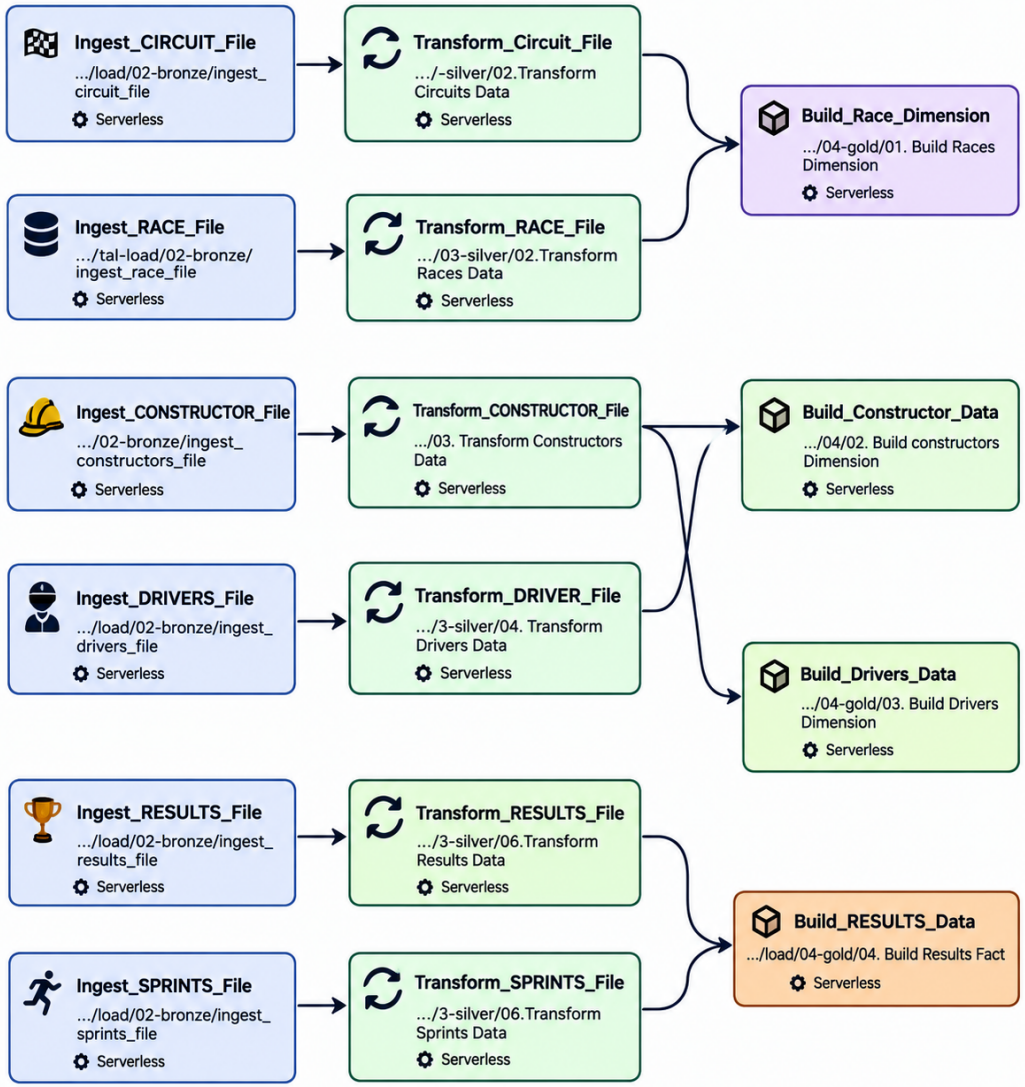
High-level overview of components, event triggers, and data movement.

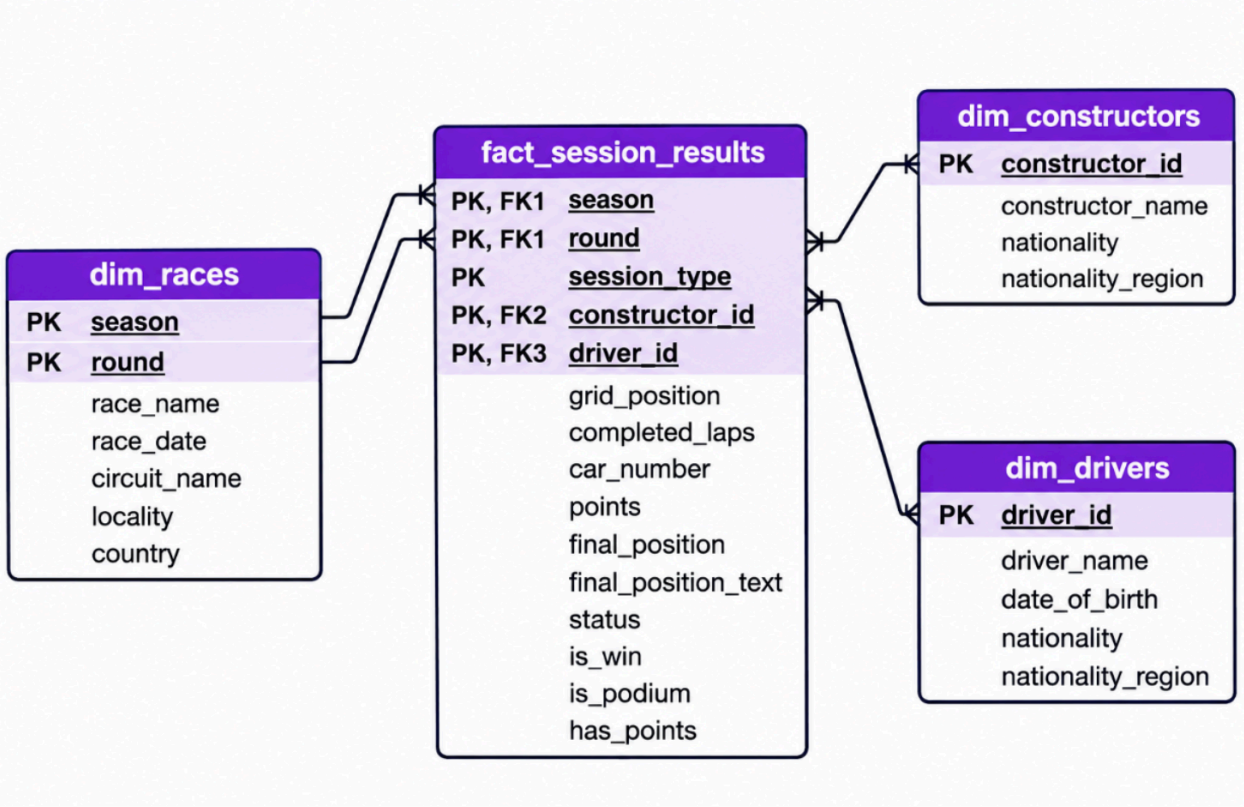
CRISP-DM + Medallion architecture & automated execution

STEP 0	PROFILE	CLEAN	DESCRIBE	MODEL
Extract	Quality	Pipeline	FAIR	Graph
25,907	29	55/55	93%	30,141
Raw rows unchanged	Issues in 7 dimensions	Validation gates passed	RDA maturity score	Typed relationships



Data Architecture : Pipeline for Dashboard and Schema





SECTION 03

FAIR Assessment

Comparative FAIR Assessment and decisions.

Same archive, two homes: 42.3% on GitHub, 84.6% on Zenodo

F-UJI 4.0.1 with metrics v0.8 (as on f-uji.net) on the four locations of the data; the Zenodo record has a DOI.

Group Zenodo record DOI, published 16 Oct 2026	**84.6%** F 86 A 86 I 67 R 100
Kaggle mirror public copy of Ergast	**84.6%** F 57 A 100 I 100 R 83
Group GitHub repository raw zip, no identifier	**42.3%** F 43 A 43 I 33 R 50
Jolpica API where the data was downloaded	**11.5%** F 14 A 29 I 0 R 0

54% → **93%**

RDA maturity over 41 indicators: as obtained → our package

6/20 → **12/20**

essential indicators met

Findable stays at RDA level 0 until the curated package has its own DOI; the raw archive already has one.

F-UJI checks machine-readable signals only: it scores a licence as present, not as compatible. The RDA model also weighs licences and provenance.

Eliminating Licensing ambiguity and metadata gaps

Five places publish the same results under four different terms. The data must stay CC BY-NC-SA 4.0.

Ergast (original source)	non-commercial use only	Origin
Jolpica-F1 (our download)	CC BY-NC-SA 4.0	Binding
Kaggle mirror	claims CC0, public domain	Conflict
Group GitHub repository	MIT licence for everything	Conflict
Group Zenodo record	CC BY 4.0	Correct it

Resolution: data CC BY-NC-SA 4.0 with attribution, code MIT.
Zenodo licence to be corrected to CC BY-NC-SA 4.0: a metadata edit, the DOI does not change.

Persistent identifiers

10.5281/zenodo.23416916

DOI of the raw archive, byte-identical to our source (MD5 checked), cited as IsDerivedFrom

98.7%

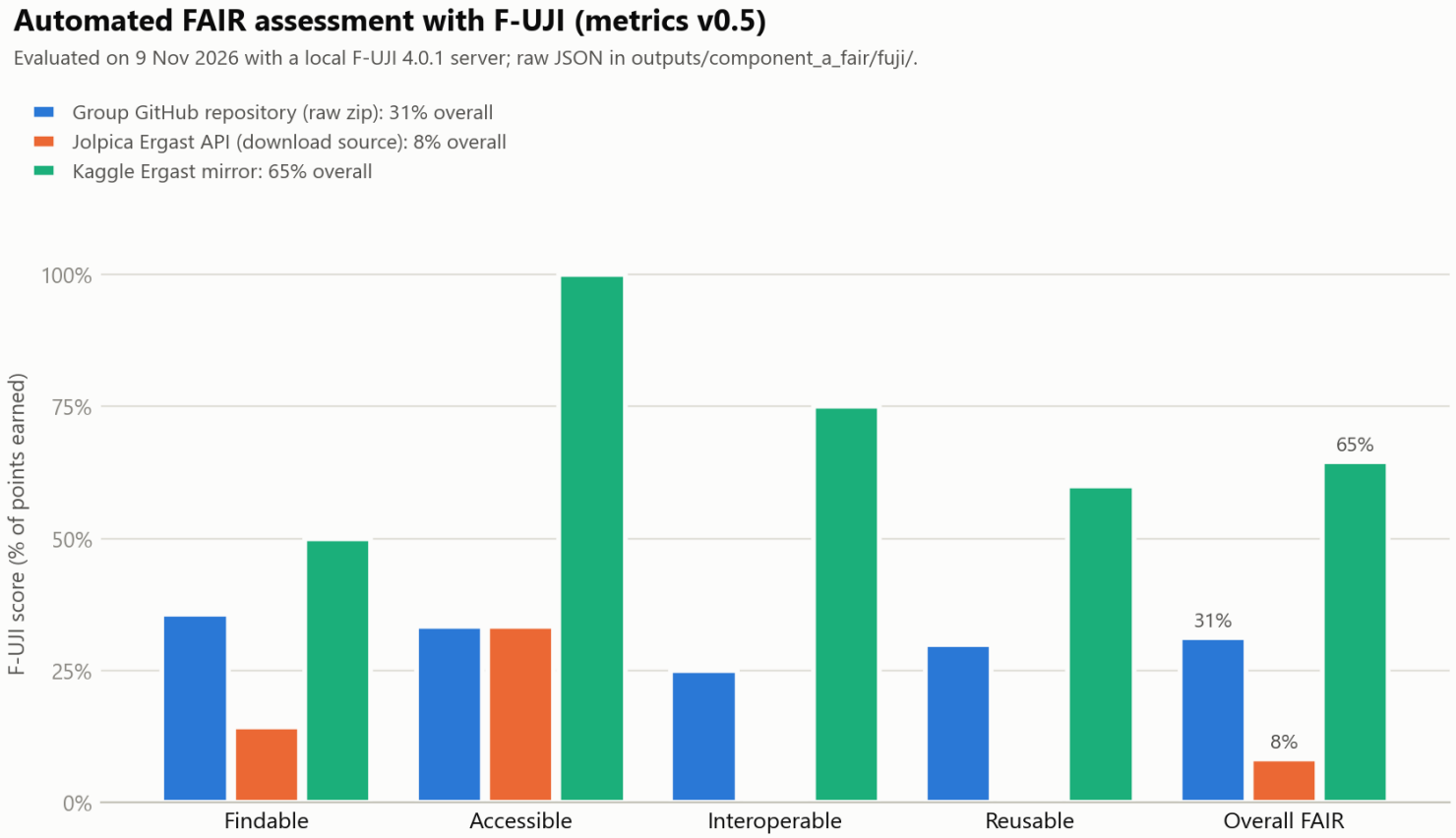
drivers linked to Wikidata (constructors 98.1%, circuits 100%)

4.5

DataCite record validated against the official XSD

Next own DOI for the curated package, ORCID IDs, ROR

As obtained, the data is hard to find and reuse



54% → 93%

RDA maturity, 41 indicators: as obtained → our package

6/20 → 12/20

essential indicators met

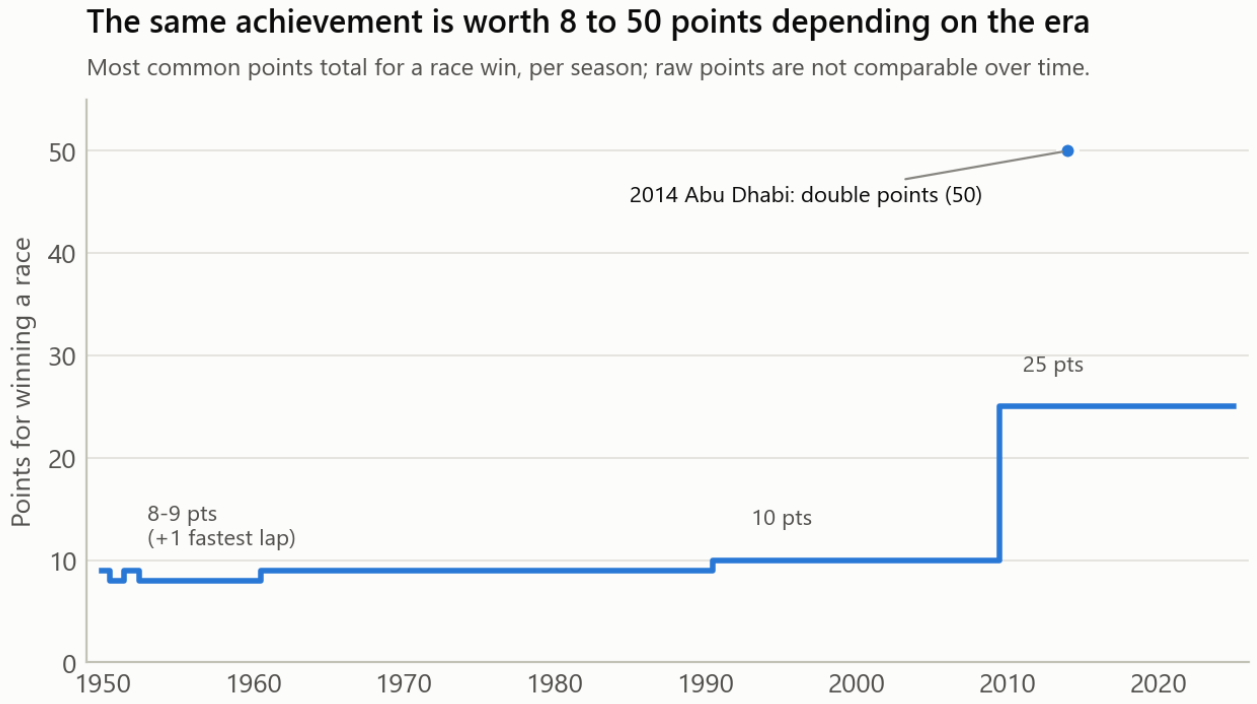
Findable stays at level 0 until a DOI is actually minted.

SECTION 02

Data Quality Verification

Schema enforcement, automated validation checks, and error isolation.

Valid history can look like bad data



8–50
points for the same achievement: winning a race

270
lap 'outliers', all valid Indianapolis 500 results: flag, never delete

Source: Ergast/Jolpica F1 results 1950-2025 (racing_f1_data.zip); group analysis.

Impact on data science and AI · Inflated wins from duplicated rows · Biased targets: a retirement as '18th' · Era bias in all-time rankings

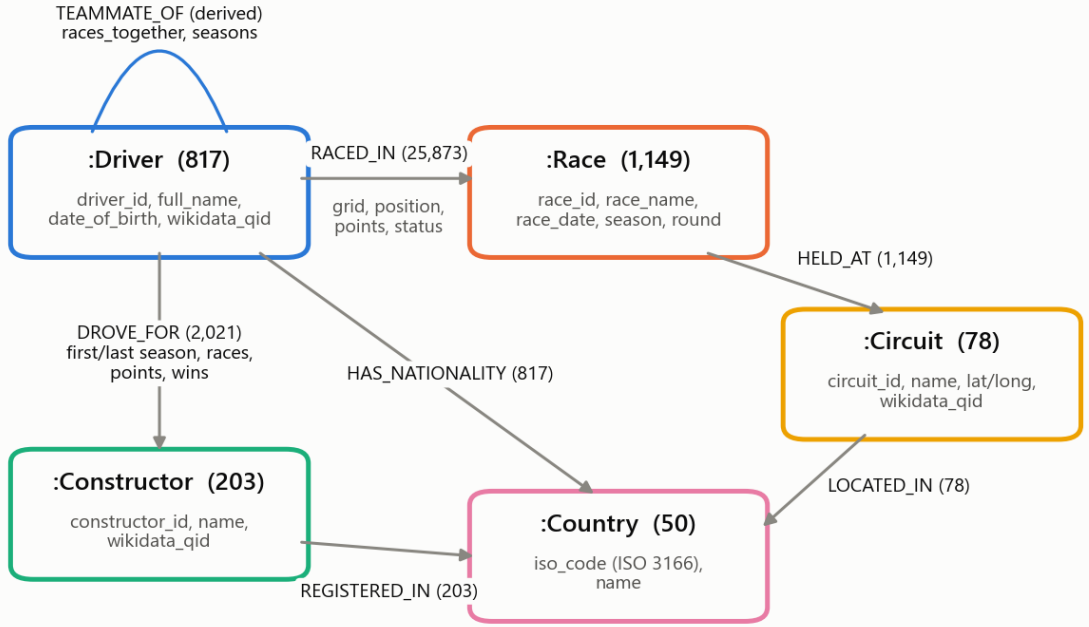
F-1 Data Quality Report

File / Dataset Name	Dimensions	Quality Dimensions	Profiling Techniques	Key Findings
Constructors	7 Cols × 211 Rows	Completeness, Uniqueness	Schema Enforcement, Null Checks	100% non-null primary keys; standardized timestamps.
Drivers	8 Cols × 860 Rows	Completeness, Referential Integrity	Null scanning, Type verification	Unique driver_id; uniform string casing & nationality mapping.
Races / Circuits	11 Cols × 1,125 Rows	Temporal Validity, Range Integrity	Date range checks, Integer bounds	Valid historical F1 bounds; zero out-of-range race dates.
Results	23 Cols × 26,080 Rows	Accuracy, Range Validation	Statistical distribution checks	Positive integer ranges verified; fully populated status codes.
Nationality Regions	5 Cols × 145 Rows	Uniqueness, Completeness	Tokenization & cardinality	100% primary key coverage; zero duplicate mapping entries.

75 years of F1 form a small world

Property graph model: 5 node labels, 7 relationship types

Counts are instances in the NetworkX graph built from the cleaned data.



Source: cleaned F1 results 1950-2025 (Component C output); NetworkX analysis.

NetworkX MultiDiGraph: 2,297 nodes, 30,141 typed edges, exported to GEXF for Gephi

0.41
clustering (random graph: 0.014)

4.8
team-mate links between two drivers on average

Fangio → Verstappen in 8 steps

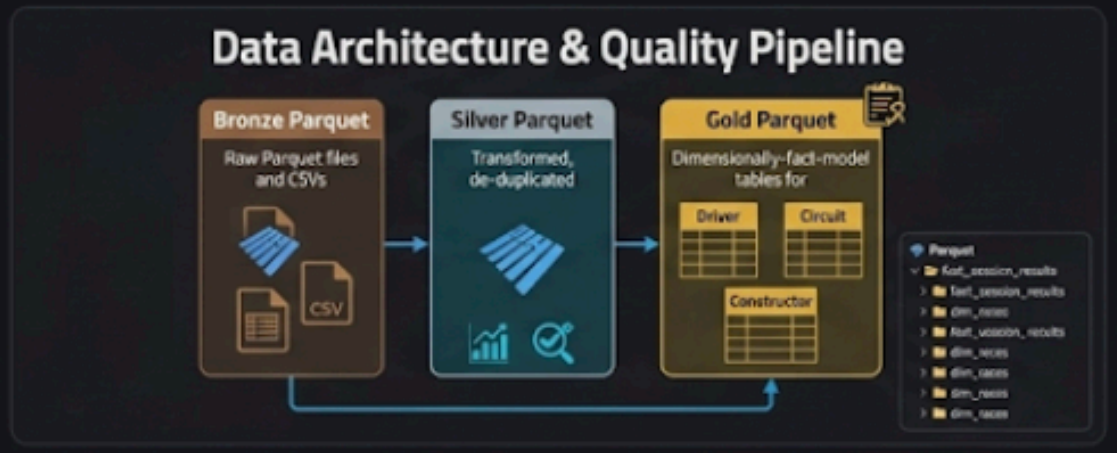
Fangio → Bonnier → Andretti → Giacomelli → de Cesaris → Barrichello → Button → Pérez → Verstappen

SECTION
05

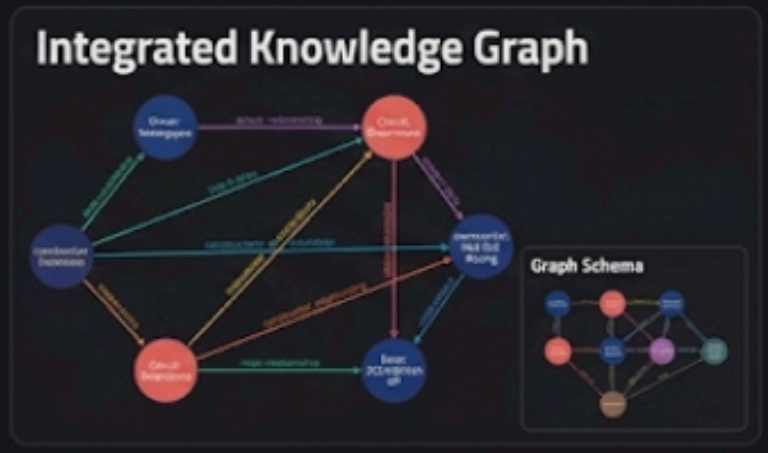
Monitoring And Ops

Latency tracking and error logging

Integrating F1 Data: From Raw Collection to Graphic-LLM-RAG Insights



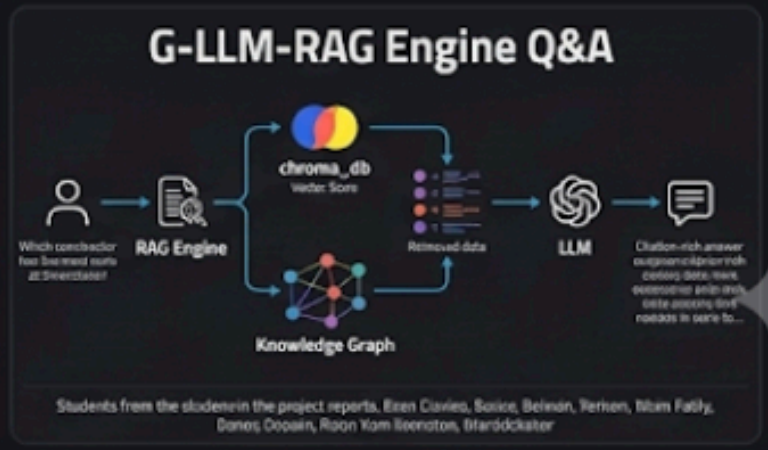
Parquet-Based Bronze-to-Gold Data Engineering with Embedded Data Validation



Neo4j Knowledge Graph & Schema for Holistic F1 Insights

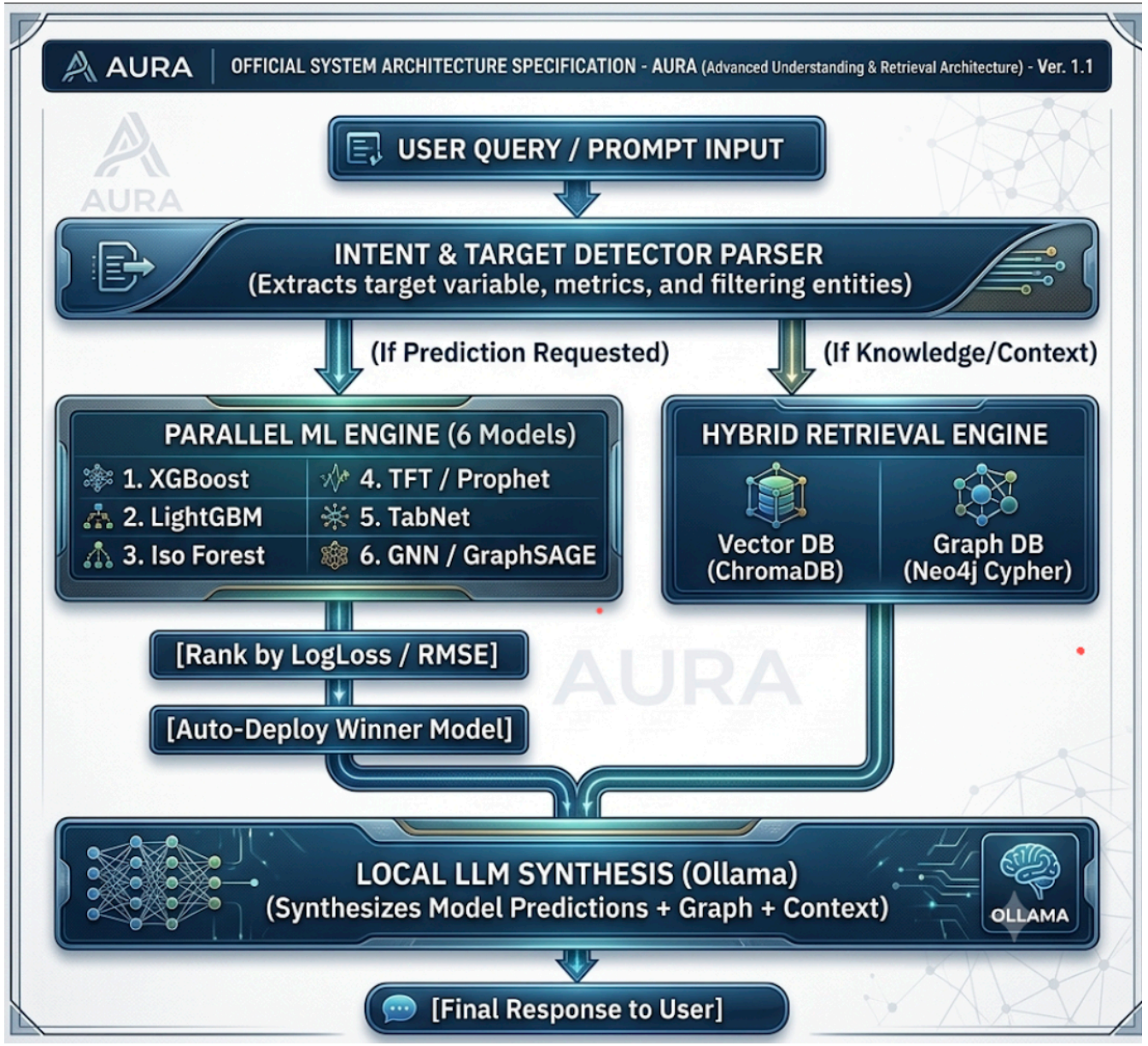


ML Engine for Anomaly Detection, Model Training, and Leaderboard Analytics



Retrieval-Augmented Generation for Data-Driven Question Answering with Citations

LLM-RAG + Neo4J Graph + Multi-Model ML Engine Architecture



What we learned

1 **Quality is contextual**

Outliers, duplicates and short grids were often valid history

2 **Order of operations matters**

11 duplicates hid behind blanked keys

3 **FAIR gaps sit around the values**

No DOI, conflicting licences, no provenance

4 **Automate the evidence**

Audit log, 55 gates, 14 tests, metadata generated from code

Questions?

Code, data, metadata and report: https://github.com/zalkimanohar/DataManagement

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